

Raj Tibarewala

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EDUCATION

Vellore Institute of Technology, Vellore

B.Tech Computer Science and Engineering (Cyber Security)

Jul 2024 – May 2028

CGPA: 8.70

EXPERIENCE

Team Ardra — Software Domain

Sept 2025 – Present

Autonomous Systems R&D

- Worked on embedded Linux platforms (Raspberry Pi, Jetson Orin Nano) for autonomous drone systems.
- Performed system setup, debugging, and validation of onboard compute modules.
- Assisted in deploying edge ML pipelines optimized for latency, power, and compute limits.

PROJECTS

SIEM-Lite — Linux Security Monitoring

Python, Streamlit, Linux

GitHub

- Built a lightweight SIEM pipeline to monitor Linux authentication logs.
- Implemented detection for SSH brute-force attacks and privilege escalation misuse.
- Designed correlation-based severity escalation for multi-stage attack detection.
- Developed a real-time Streamlit dashboard for alerts and timeline analysis.

GeoAI Hack — IIT Bombay TechFest

ML, Docker, FastAPI, Geospatial Analysis

- Trained YOLO models for rooftop and infrastructure detection from satellite imagery.
- Applied SegFormer-based semantic segmentation for continuous geospatial features.
- Deployed inference pipeline using FastAPI and Docker.

Astrophysics Data Analysis

Python, NumPy, Pandas, Matplotlib

- Estimated Hubble Constant using Type Ia supernova datasets; validated against Planck-18 results.

PUBLICATIONS

Nuclear Thermal Rocket: Design Considerations and Efficiency Optimization for Deep-Space Explorations | Paper

Co-authored under faculty supervision

- Conducted literature review and background research on nuclear thermal propulsion systems.
- Assisted in structuring technical content and supporting manuscript preparation.

TECHNICAL SKILLS

Languages: Python, C, C++, Java

Security: SIEM concepts, Linux security, log analysis, attack correlation

ML/Data: NumPy, Pandas, Scikit-learn, OpenCV, Matplotlib

Tools: Git, Docker, Linux, VS Code, Jetson, Raspberry Pi

CERTIFICATIONS

Google Cybersecurity Professional Certificate – Coursera

Astronomy and Astrophysics Summer School – India Space Academy (2025)